

Exercises

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Surname, First name**Applied Microeconomics (FEB12001X)**

FEB12001X - Resit 7 July 2026

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2	2	2	2	2	2
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General information

Lecturers: Josse Delfgaauw and Robert Dur

Date & Time of examination: 7 July 2026, 13:30 - 16:30

Type of examination: Closed book, Open questions

Number of questions: 9 questions (question 10 is additional answer space)

Number of pages: 24 pages (incl. cover sheet).

Separate appendix: no

Instructions

- Please use only a pen with black or blue ink.
- Fill in your name and student number as indicated above. **Also, code your student number by coloring the circles with the corresponding numbers.** If you made a mistake, put a cross through the wrong number, and color the correct one.
- **You have to provide a calculation or explanation of each answer.** An answer without calculation or explanation yields zero points.
- Write your answer **in** the answer box corresponding to the question.
- Because the total answer space is limited, we strongly advise you to prepare your answers on scrap paper: **no additional exam paper will be provided.**
- In case you made a mistake and therefore you need more space, use the additional writing space at the end of the exam. Clearly indicate where the (remaining part of the) answer can be found.

Permitted materials

- You are allowed to use a pen and blank scrap paper.
- You are allowed to use a calculator of **only** the following types: FX-82 / FX-85 / FX-95 / FX-100 / FX-350 / FX-570 / FX-580 / FX-991 / TI-30 / TI-34 / TI-36.
- You are NOT allowed to use notes
- You are NOT allowed to use books.
- You are NOT allowed to use a dictionary.

- the cover sheet continues on the next page -

Please note:

- Exam questions and/or answers may not be copied or disseminated in any way.
- Scrap paper and other exam related documents may not leave the exam room.
- It is not allowed to keep a watch within reach during the examination. Watches must be stored in your coat or bag.
- All mobile devices or (potential) data media must be switched off during the examination and stored in your coat or bag.

Good luck!

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Question 1 - Public economics

Suppose there are two individuals, Anne and Ben, who both derive utility from the consumption of a public good. Anne's valuation of the good in euro is given by

$$U_A = aq_A - \frac{1}{2}(q_A)^2,$$

where q_A is the quantity consumed by Anne. Ben's valuation in euro is given by

$$U_B = bq_B - \frac{1}{2}(q_B)^2,$$

where q_B is the quantity consumed by Ben. Supply of the public good on the market is perfectly elastic. Marginal production cost are equal to the price $p = MC = c$ euro. We assume that $a > b > c$ and that $b + c > a$.

- 2p **1a** Derive the socially efficient level of consumption of Anne and Ben.

Suppose that the public good is provided by the market. Anne and Ben decide independently how many units of the public good to buy (of course taking the presence of the other into account).

- 2p **1b** Determine how many units of the public good Ben consumes in the Nash-equilibrium that arises on the market.



Now suppose the government offers a subsidy per unit of production to producers of the public good. The Pigouvian level of the subsidy equals $\frac{1}{2}(b + c - a)$ euro.

- 3p **1c** Explain (in words) why the Pigouvian subsidy increases in b . Use no more than three sentences.



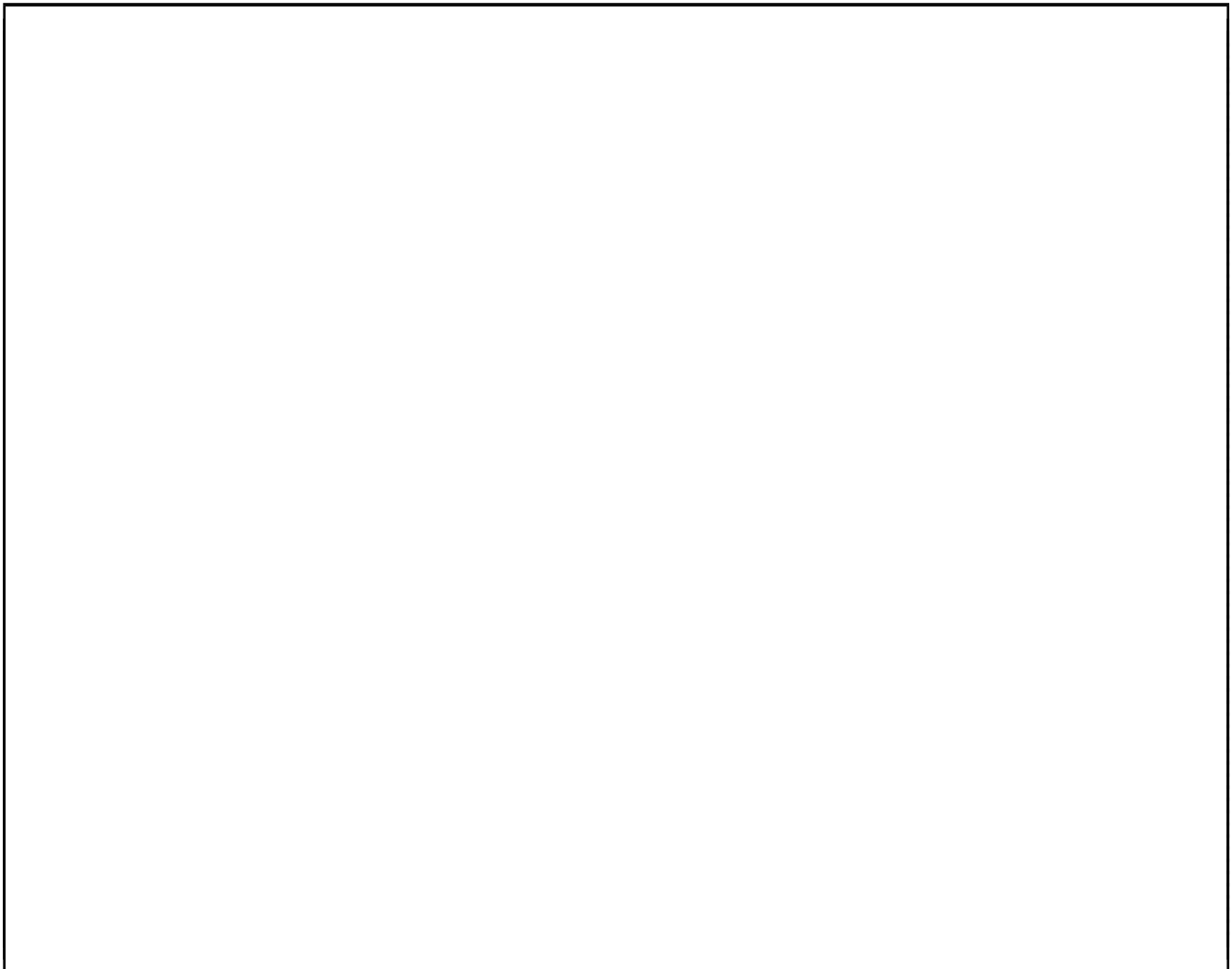
In case of pure public goods, at least one of the conditions of the Coase Theorem is always violated.

- 2p **1d** Which of the characteristics of a pure public good makes that at least one of the conditions of the Coase Theorem is always violated? Explain why. Use no more than four sentences.

Question 2 - Public economics

Bert has 200 hours per month which he has to divide between working and leisure. His net wage is 20 euro per hour. Leisure L is the number of hours (out of the 200) that he does not spend working (hence, $L = 200 - \text{number_hours_work}$). The government has introduced the following system of poverty alleviation: People who have less than 500 euro per month earnings from work receive a grant of 600 euro per month. For people who earn more than 500 euro per month, the grant is reduced by 50 cents for every euro earnings from work above 500 euro. People who earn 1700 euro or more from work receive no grant. You can ignore taxes throughout this exercise.

- 3p **2a** Draw Bert's budget constraint, with total net income on the Y-axis and number of hours of leisure on the X-axis. Clearly indicate the relevant points on the budget constraint on both axes (with numbers).



Bert has diminishing marginal utility of both income and leisure (such that his indifference curves relating to the trade-off between income and leisure have the normal shape (convex)). Suppose that if the government would not have a system of poverty alleviation (grant is 0 for everyone), Bert would choose to work 85 hours ($L=115$).

- 2p **2b** Argue whether Bert works more than 85 hours, less than 85 hours, or exactly 85 hours given that the government has introduced the poverty alleviation system as described above. Argumentation is crucial. (Hint: Calculations are neither necessary nor possible. You can use a new sketch of Bert's budget constraint in your argumentation.)

- 2p **2c** Suppose you would know Bert's utility function. Then, it would be possible to calculate the loss of efficiency due to Bert's response to the introduction of this poverty alleviation system. What steps must you take to calculate this? Note: you do not need to perform the calculation. It is not sufficient to mention terms; you must clearly indicate the steps required to carry out this calculation.



**Question 3 - Public economics**

Consider the market for good X . Demand for good X is given by

$$P = 200 - \frac{1}{2}Q_D$$

where P is the price of good X (in euro) and Q_D is quantity demanded. Supply of good X is given by

$$P = 20 + Q_S$$

where Q_S is quantity supplied. Consumption of good X leads to external cost. The marginal external cost are given by

$$MEC = \frac{1}{2}Q_D$$

Suppose there are many identical consumers of good X . Furthermore, suppose that the external cost of the consumption of good X are fully incurred by consumers of good X .

- 2p **3a** Explain (in words) why market failure arises on the market for good X , even if the people that are hurt by the external cost are consumers of good X . Use no more than two sentences.



- 2p **3b** Determine the (economic) tax incidence of a tax of 30 euro per unit levied on the consumers of good X .

- 2p **3c** Explain the effect of a higher elasticity of supply of good X on the tax incidence. Use no more than two sentences.

- 2p **3d** Determine whether a unit tax of 30 euro increases or decreases efficiency on the market for good X , compared to the situation without a tax on good X .





2p **3e** Determine the level of a Pigouvian tax on good X .

Question 4 - Public economics

Suppose that the citizens of a country differ in productivity α . This productivity α is uniformly distributed between 0 and 1. A citizen with productivity α who has a job earns income $Y = 2500\alpha$ euro. A citizen without a job earns income $Y = 225$ euro. Utility of income is given by $U = \sqrt{Y}$. Searching for a job is difficult and time-consuming, and yields a loss in utility equal to c . If a citizen searches for a job, the probability that the citizen finds a job is given by π . A citizen who does not search never gets a job. Citizens maximise their expected utility.

The government considers introducing the following system of poverty alleviation. Citizens who earn less than 625 euro receive a benefit that increases their total income to 625 euro (such that the benefit equals $625 - \text{own_earnings}$). The government cannot observe an individual's productivity α , but can observe earnings and whether citizens search for a job.

Throughout the exercise, you can ignore taxes.



- 2p **4a** Suppose $\pi = 0.8$ and $c = 8$. Determine which citizens search for a job before the introduction of the benefit system.

The introduction of the system of poverty alleviation may reduce efficiency, because fewer citizens search for a job. A politician proposes to give the benefit only to citizens that engage in job search. In other words, people only get the benefit if they searched for a job and earn less than 625 euro (either because they did not find a job or because their job pays less than 625 euro). A citizen who did not search for a job, does not receive any benefit (implying that the citizen has an income of 225 euro).

- 1p **4b** Explain (in words) why the politician's proposal to make job search mandatory for receiving benefits could **reduce** efficiency. Use no more than three sentences.



- 1p **4c** Suppose the politician's proposal to make job search mandatory for receiving benefits **increases** efficiency. From a normative perspective, why could it still be optimal to **not** make job search mandatory? Use no more than two sentences.

Question 5 - Personnel economics

Astrid and Bella are sisters. Each of them has a job in which she can freely decide on her work hours. h_A denotes the number of hours worked by Astrid and h_B denotes the number hours worked by Bella. They earn w per hour. Astrid's utility function is:

$$wh_A + \mu(wh_A - wh_B) - (h_A)^2,$$

where the first term reflects utility from income and the last term reflects the utility costs from loss of leisure. The term in the middle reflects that Astrid finds it important how her income compares to the income of her sister. We assume that $\mu > 0$, and so assume that Astrid likes it when she earns more than her sister and dislikes it to earn less than her. The utility function of Bella is $wh_B + \mu(wh_B - wh_A) - (h_B)^2$. The sisters choose their hours independently from each other and simultaneously.

- 2p **5a** How many hours will Astrid and Bella choose to work?

- 2p **5b** Assume now that Astrid and Bella jointly decide on their work hours. They choose work hours so as to maximize the sum of their utilities. How many hours will Astrid and Bella work?

- 2p **5c** Now assume that Astrid and Bella can not jointly choose their work hours, but can jointly decide to implement an income tax scheme. Any tax receipts will be donated to a charity that Astrid and Bella do not care about. The income tax system can be as simple or complex as they like. Do Astrid and Bella jointly decide to implement a income tax scheme and, if yes, how does it look like? Motivate your answer.



**Question 6 - Personnel economics**

A company employs one worker. The worker has the following utility function:

$$U = bq + \gamma e - \frac{1}{2}\theta e^2$$

where U is the utility of the worker, $b \geq 0$ is the bonus per unit of output, q is the level of output, e the level of effort of the worker, $\gamma > 0$ and $\theta > 0$. The parameter γ can be interpreted as a measure of intrinsic motivation of the worker. The worker aims for maximum utility. The production function is $q = e$. The company sells the output at a price of $p > 0$ per unit of output. We assume that $p > \gamma$. Throughout this exercise, we assume that the worker is better off with this job than with any other job or without a job. The company does not pay a base salary to the worker. The profit of the company is the difference between sales revenues and bonus costs. The company aims for maximum profit. Social welfare is defined as the sum of the utility of the worker and the profit of the company.

2p **6a** Derive the level of effort that maximizes social welfare.



- 2p **6b** Derive whether in the above model the following statement is true: If the worker has a higher intrinsic motivation (that is, if γ is higher), then an increase of the bonus has a smaller effect on the level of effort that is chosen by the worker.

- 4p **6c** Derive what level of the bonus the company chooses. Note: You can get four points for this!



- 2p **6d** Explain in words why the profit-maximizing level of the bonus depends on the level of intrinsic motivation γ . Note: we ask for economic reasoning here, not for a mathematical derivation or mathematical reasoning. Use at most three sentences.



Question 7 - Personnel economics

A municipality is searching for workers to produce public goods. In total, the municipality would like to reach a production of size Q . There are two types of workers in the labor market: motivated and unmotivated workers. Motivated workers have the following utility function:

$$U = w + \gamma e - \theta e^2$$

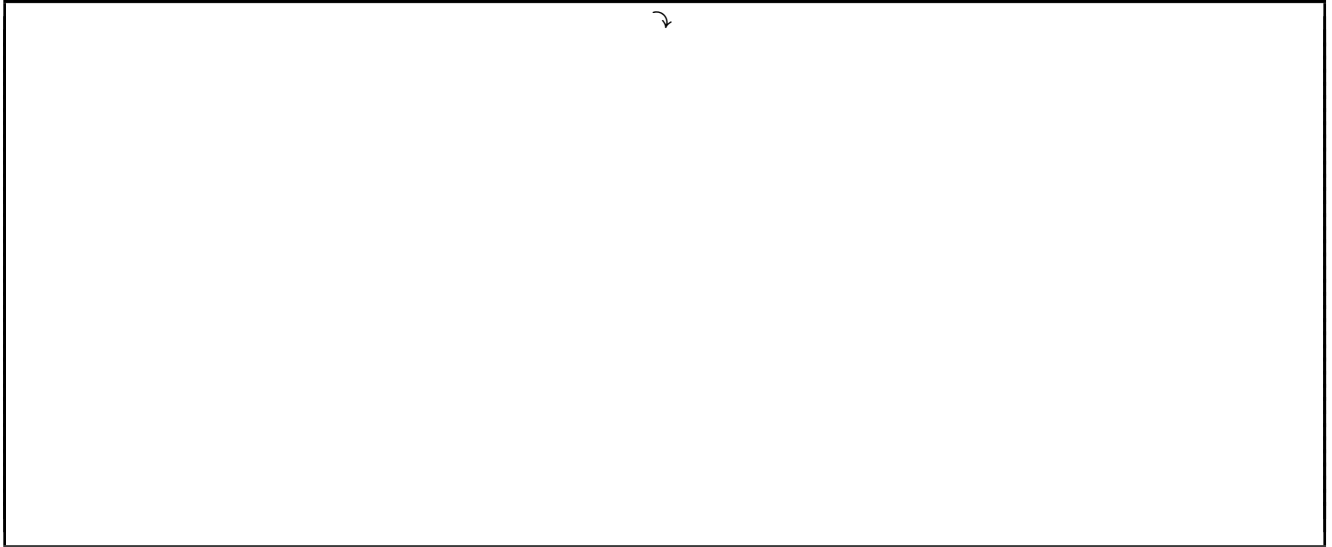
where U is utility, w the base salary, e the level of effort, and $\gamma > 0$ a positive parameter. Unmotivated workers have the following utility function:

$$U = w - \theta e^2$$

We assume that the municipality cannot pay any bonuses. If a worker decides not to work at the municipality, then he or she achieves utility level U^{alt} . This holds for both the motivated and unmotivated workers. The production level achieved by a worker is equal to his or her effort level. Finally, we assume that if somebody applies at the municipality, the municipality can observe whether it is a motivated or unmotivated worker. There is a large number of both types of workers present in the labor market, and so if the municipality wants to use workers of one single type, then there are sufficiently many available to produce production Q .

- 1p **7a** Calculate how much production an unmotivated worker produces if he is hired by the municipality.

- 1p **7b** Show that if the parameter γ is higher, that then a motivated worker chooses a higher level of effort and is willing to work for a lower base salary.

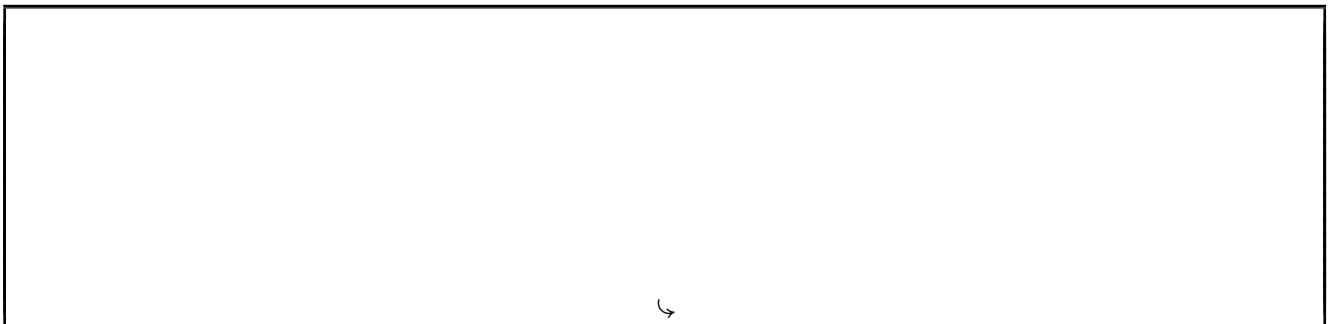


1p **7c** How many motivated workers are needed to achieve production level Q ?



Assume now that the municipality cannot distinguish between the two types of workers during the application procedure.

1p **7d** Does this lack of information cause a problem for the municipality? If not, explain why not. If yes, then explain what problem arises.



**Question 8 - Personnel economics**

The HR manager of a large retail company consisting of 900 stores wonders whether the personnel policies can be improved. Currently, store managers have a lot of autonomy in choosing the personnel policies in their store. The HR manager collected data and found out that 400 store managers currently use performance pay for workers whereas the other 500 store managers do not.

- 2p **8a** A personnel economist argues that, according to theory, stores that use performance pay for workers may achieve higher sales and profits for at least two reasons, one related to moral hazard and another related to adverse selection. Explain to the HR manager in two clear sentences what the personnel economist tries to tell.



- 2p **8b** A data scientist had a look at the sales and profits data and found out that the 400 stores that use performance pay for workers do significantly worse than the 500 stores that do not. Give one possible reason for this pattern in the data. Use at most two sentences. You can use technical terms, but you need to explain them.

- 2p **8c** The personnel economist proposes to run a field experiment to find out whether sales and profits would increase if performance pay for workers was abandoned in the 400 stores that use them. Sketch in at most three sentences how such a field experiment should be designed so as to provide credible evidence of the effect of performance pay for workers on sales and profits.



Question 9 - Personnel economics

A not-for-profit organization aims to maximize output. To produce output, the organization hires workers. Each worker contributes q to total output. If the organization offers wage W , then the organization hires $N(W)$ workers, where $N'(W) > 0$. The not-for-profit has a budget equal to B .

- 2p **9a** Derive by how much the total expenditures of the organization increase when it offers a one euro higher wage.

- 2p **9b** Under what condition does this organization classify as an organization with monopsony power?



Additional answer space

Only use this extra space if you made a mistake in one of your answers. Please do not forget to **indicate in the answer box of that particular question where the correct answer can be found**. Otherwise, the answers in the additional space below will not be reviewed.

NB: Do **not** use this additional space for answers that are too long.

10a Additional answer box 1



10b Additional answer box 2

10c Additional answer box 3



10d Additional answer box 4

10e Additional answer box 5

